

5A The language of algebra

Learning Objective	Apply algebraic properties to rearrange, expand and factorise linear expressions
Today We Are	Identifying the basic terminology of algebra
Indicators of Progress	- I can identify coefficients, terms, constant terms in expressions - I can write expressions from word descriptions

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READY FOR A CHALLENGE?

Use your math skills to find the value of each '?'

 +
  +
  = 21

10 =
  +
 4 +
 

 +
  = 8

14 =
  +
 

 +
  +
  +
  = ?

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Strawberry : 3
 orange : 5
 coconut : 9
 watermelon : 3
 ? : 20

Key Ideas

- In algebra, letters can be used to represent one or more numbers. These letters are called **pronumerals** or **variables**.
- $a \times b$ is written ab and $a \div b$ is written $\frac{a}{b}$.
- $a \times a$ is written a^2 .
- An **expression** is a combination of numbers and pronumerals combined with mathematical operations; for example, $3x + 2yz$ and $8 \div (3a - 2b) + 41$ are expressions.
- A **term** is a part of an expression with only pronumerals, numbers, multiplication and division; for example, $9a$, $10cd$ and $\frac{3x}{5}$ are all terms.
- A **coefficient** is the number in front of a pronumeral. If the term is being subtracted, the coefficient is a negative number, and if there is no number in front, the coefficient is 1. For the expression $3x + y - 7z$, the coefficient of x is 3, the coefficient of y is 1 and the coefficient of z is -7 .
- A term that does not contain any variables is called a **constant term**.
- The **sum** of a and b is $a + b$.

20

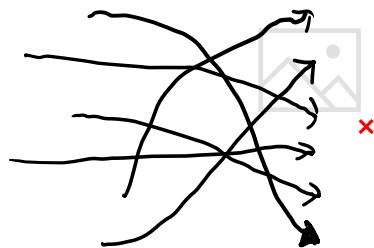
— The **sum** of a and b is $a + b$.

■ The **difference** of a and b is $a - b$.

■ The **product** of a and b is $a \times b$.

■ The **quotient** of a and b is $a \div b$.

■ The **square** of a is a^2 .



a) $4a, b, -12c, 5$

b) $a = 4, b = 1, c = -12, d = 0$

c) 5

d) 7

$$\begin{array}{l}
 \text{R} \rightarrow \\
 m \times 7 \text{ or } 7m \times \\
 \frac{1}{2}h + 5 \text{ or } \frac{h}{2} + 5 \\
 (a+b) \times 2 \text{ or } 2(a+b)
 \end{array}$$



x



x



x



x

